WOOJIN LEE

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Computational chemist with a Ph.D. and 5+ years of experience building high-performance computing (HPC) workflows for drug discovery, structural biology, and computational organic chemistry. Skilled in accelerating large-scale scientific workloads across distributed computing environments, including giga-scale virtual screening of 3 trillion compounds and GPU-optimized molecular dynamics (MD) simulations using GROMACS. Adept at full-stack scientific programming (Python, Linux) and software (Gaussian, ORCA) to build robust molecular modeling, cheminformatics, and conformational analysis pipelines. Demonstrated success as a computational chemist, with a strong publication record and a history of collaborating with cross-functional experimental teams to translate complex computational results into actionable experimental strategies.

SKILLS

- **Computational Chemistry and Biology**: Molecular dynamics (GROMACS, Progdyn Suite), quantum mechanical simulations (Gaussian, ORCA), and molecular docking (Molsoft ICM-Pro, AutoDock Vina), and structural modeling (Robetta, AlphaFold, RosettaCM)
- **Computer-Aided Drug Design (CADD):** Giga-scale virtual ligand screening, structure-based drug discovery, early-stage hit identification and lead optimization (Molsoft ICM-Pro, V-SYNTHES, GROMACS, RDKit)
- **Data Analysis & Cheminformatics:** Python, Jupyter Notebook, RDKit, QSAR analysis using molecular descriptors, and molecular dynamics trajectory analysis (MDTraj, MDAnalysis)
- **Conformational & Energetic Analysis:** Conformational ensembles and relative energy calculations (Molsoft ICM-Pro, Spartan 18, XTB-CREST)
- **High-Performance Computing (HPC):** Linux-based HPC environments, parallelized workflows, and resource management (UCLA IDRE, NSF XSEDE, USC CARC)

WORK EXPERIENCE

UNIVERSITY OF SOUTHERN CALIFORNIA Los Angeles, CA

Postdoctoral Research Associate in Prof. Vsevolod "Seva" Katritch's research group Oct 2023 – Present

- Conducted structure-based drug discovery for prostaglandin D_2 receptor 1 (DP1) by integrating molecular docking, all-atom MD simulations, and structural biology to define ligand recognition and activation mechanisms
- Led giga-scale virtual screening of ~3 trillion Enamine REAL make-on-demand compounds using the V-SYNTHES platform and prioritized 269 candidates for synthesis and *in vitro* functional evaluation
- Enabled the discovery of 23 novel DP1 modulators (20 antagonists and 3 agonists); 209 prioritized compounds were synthesized, with best potencies of IC_{50} = 1.1 μM and EC_{50} = 1.3 μM
- Contributed to structural and mechanistic characterization of DP1 by identifying key features: determinants of receptor modulation, a DP1-specific activation switch in the sodium allosteric pocket, and a stabilizing TM6–helix 8 interaction network
- Constructed all-atom 3D models of the human M_2R-β-arrestin-1 complex using RosettaCM and ICM-Pro, translating 65 experimental crosslinking pairs into harmonic restraints to resolve the uncharacterized ICL3 interface
- Performed 36 μs of unbiased MD simulations in GROMACS to evaluate global complex stability and validate the dynamic accessibility of the binding interface
- Elucidated a binding topology in which the phosphorylated ICL3 wraps around the arrestin N-domain while maintaining a dynamic core engagement
- Collaborated with Prof. Vadim Cherezov (University of Southern California) and Prof. Irene Coin (Leipzig University) to integrate computational, structural, and pharmacological studies across DP1 and M_2R-β-arrestin-1 projects

UNIVERSITY OF CALIFORNIA, LOS ANGELES Los Angeles, CA

Graduate Research Assistant Oct 2018 – Oct 2023

- Elucidated the behavior, reactivity, and full reaction pathways of short-lived carbocation and radical intermediates by applying Density Functional Theory (DFT) and MD simulations (Progdyn)
- Discovered critical mechanistic insights, including rate-determining steps, divergent reaction pathways, and mechanistic switches in complex reaction systems
- Collaborated with international experimental laboratories (Brandeis University, Dalian University of Technology, University of Rome, University of British Columbia) to validate computational findings, resulting in robust mechanistic conclusions and multiple high-impact publications

Graduate Teaching Assistant Oct 2018 – Dec 2022

- Delivered organic chemistry instruction and laboratory training to 200+ undergraduate students, demonstrating strong communication skills to break down complex chemical concepts and provide strong mentorship

CALIFORNIA INSTITUTE OF TECHNOLOGY Pasadena, CA

Visiting Student Researcher Oct 2021 – Sep 2022

- Modeled complex organic reaction mechanisms, discovering novel cationic [3,3] and [1,3] rearrangements and elucidating Friedel-Crafts pathways in the formation of medium-sized rings

GENENTECH, INC. South San Francisco, CA

Medicinal Chemist Intern in the Small Molecule Discovery Chemistry Department May 2017 – Sep 2017

- Streamlined synthetic routes and optimized yields for dendrimer drug conjugates while gaining industry experience in small-molecule drug discovery workflows

UNIVERSITY OF CALIFORNIA, BERKELEY Berkeley, CA

Research Assistant in Prof. F. Dean Toste's research group Jan 2016 – May 2017

- Developed extensive hands-on experience in synthetic organic chemistry, including reaction design, synthesis, purification, and characterization
- Designed and executed experimental studies to optimize the product yield and investigate reaction mechanisms

SELECTED PUBLICATIONS

Full publication list available on Google Scholar

[3] Müller, T.[#]; **Lee, W.**[#]; Alkara, A.; Baischew, A.; Aydin, Y.; Lam, J. H.; Nagel, F.; Hoffmann, C.; Schultz, S.; Sinz, A.; Katritch, V.; Coin, I. "Phosphorylation site topology governs the functional dynamics of arrestin recruitment to GPCRs," *Manuscript submitted* [[#]*Authors contributed equally*]

[2] Davoudinasab, B.; Raskovalov, A.; **Lee, W.**; Kim, D.; Kim, H.; Lam, J. H.; Han, G. W.; Katritch, V.; Cherezov, V. "Structural Insights into the Mechanism of Activation and Inhibition of the Prostaglandin D_2 receptor 1," *Nat. Commun*., **2025**, *16,* 8944.

[1] **Lee, W.**; Benton, T.; Sengupta, A.; Houk, K. N. "Molecular Dynamics of the Norbornyl Cation and Its Generation in Winstein–Trifan Solvolysis: The Timing of Sigma-Bridging," *J. Org. Chem.,* **2024**, *89*, 1140–1146.
[*Highlighted in ACS Editors' Choice, UCLA Chemistry & Biochemistry Newsroom, and CNSI Newsroom*]

PRESENTATIONS

THE FOURTH DRUG DISCOVERY INNOVATION WORKSHOP Los Angeles, CA

University of Southern California Feb 2026

Title: Virtual Ligand Screening and Molecular Dynamics Simulations of the Prostaglandin D_2 Receptor 1

NATIONAL ORGANIC CHEMISTRY SYMPOSIUM POSTER PRESENTATION San Diego, CA

American Chemical Society Division of Organic Chemistry June 2022

Title: Diastereoselective Radical Aminoacylation of Olefins through N-Heterocyclic Carbene Catalysis

AWARDS

- Hanwha TotalEnergies Industry-Academic Scholarship Nominee (~$45,000) 2023
- Excellence in Chemistry Fellowship (>$300,000) 2018 – 2023
  (5-year fellowship covering full international tuition, health insurance, and living stipend)
- UCLA Doctoral Student Travel Grant ($1,000) 2022
- Graduate Dean's Scholar Award ($14,500) 2018

EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES Los Angeles, CA

Doctor of Philosophy, Chemistry

- Research Advisors: Prof. Kendall N. Houk and Prof. Hosea M. Nelson
- Graduated: September 2023
- Thesis: Exploring Reaction Mechanisms of Short-Lived Carbocation and Radical Intermediates via Synergetic Modern Computational Methods and Collaborative Experiments

UNIVERSITY OF CALIFORNIA, BERKELEY Berkeley, CA

Bachelor of Science, Chemistry

- Research Advisor: Prof. F. Dean Toste
- Graduated: May 2016